



Solve all the question in this paper for o level chemistry question .
show the question together with the answer and explanation

Vot

Topia

Mole Concept and Stoichiometry

Multiple Choice Questions

1. What is the relative molecular mass, M_r , of $(\text{NH}_4)_2\text{CO}_3$?

$$(14 \times 2) + (1 \times 8) + (12 + (16 \times 3))$$

A 77

B

80

C

94

D

96

$$26 + 8 + 12 + 48 \text{ [N23/1/26]}$$

2. Acid rain contains sulfuric acid

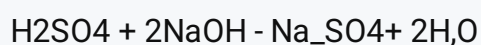
1 96

25 cm³ of acid rain is neutralised by 28 cm³ of 0.5 mol/dm³ aqueous sodium hydroxide.

The equation for the reaction is shown.

25.28

A



What is the concentration of sulfuric acid in the acid rain?

A

0.28 mol/dm³

B

0.56 mol/dm³

C

1.12 mol/dm³

D

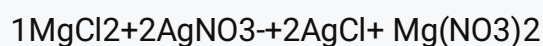
2.80 mol/dm³

(A -)

[N23/1/27]

3. A solution containing 3.80 g of magnesium chloride is reacted with an excess of aqueous silver nitrate.

3.80



mo1= 3.80

24 +(35.5×2)

What is the mass of silver chloride produced?

A 2.87 g

B

4.98 g

C 5.74g

D

11.48 g

(D)

muss =molxm-

[N22/1/6]

0,08 x (10443s.5]

4. 5.85 g of sodium chloride, NaCl, is dissolved in a solution with a total volume of 100 cm³.

=11.48 g

90 cm³ of this solution is poured into a large beaker and the remaining 10 cm³ into a small beaker.

90 = 0.09

5.265

0.265702

234 39.5

58.5

5.457565

5.85g NaCl

90 cm³

10 cm³

in 100 cm³

4

Which statement is correct?

A

The concentration of NaCl in the large beaker is 0.1 mol/dm³.

B

The concentration of NaCl in the large beaker is 52.65 g/dm³

C

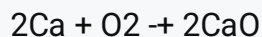
The concentration of NaCl in the small beaker is 0.585 g/dm³.

D

The concentration of NaCl in the small beaker is 1 mol/dm³.

[N22/1/7]

5. 2.0 g of calcium is completely burnt in pure oxygen.



Which volume of oxygen is used in this reaction at room temperature and pressure?

A 0.025 dm³

B 0.050 dm³

C 0.60 dm³

D 1.2 dm³

(C)

[N21/1/26]

O Level Science Chemistry

Topic 7

Mole Concept and Stoichiometry

25



Here are the answers, explanations, and the original questions for the **O Level Chemistry (Mole Concept and Stoichiometry)** multiple-choice questions you provided.

Question 1

What is the relative molecular mass, M_r , of $(\text{NH}_4)_2\text{CO}_3$?

A 77

B 80

C 94

D 96

Answer: D (96)

Explanation:

$(\text{NH}_4)_2\text{CO}_3$ contains:

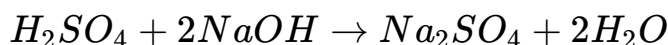
- N: 2 atoms $\rightarrow 2 \times 14 = 28$

- H: 8 atoms $\rightarrow 8 \times 1 = 8$
- C: 1 atom $\rightarrow 1 \times 12 = 12$
- O: 3 atoms $\rightarrow 3 \times 16 = 48$

$$M_r = 28 + 8 + 12 + 48 = 96$$

Question 2

25 cm³ of acid rain is neutralised by 28 cm³ of 0.5 mol/dm³ aqueous sodium hydroxide.
The equation is:



What is the concentration of sulfuric acid in the acid rain?

- A 0.28 mol/dm³
- B 0.56 mol/dm³
- C 1.12 mol/dm³
- D 2.80 mol/dm³

Answer: A (0.28 mol/dm³)

Explanation:

1. Moles of NaOH:

$$\text{Volume} = 28 \text{ cm}^3 = 0.028 \text{ dm}^3$$

$$\text{Moles} = 0.028 \times 0.5 = 0.014 \text{ mol}$$

2. From the equation: 1 mol H_2SO_4 reacts with 2 mol NaOH.

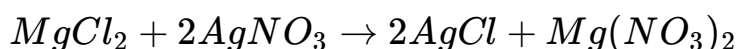
$$\text{Moles of } H_2SO_4 = \frac{0.014}{2} = 0.007 \text{ mol}$$

3. Volume of acid rain = 25 cm³ = 0.025 dm³.

$$\text{Concentration} = \frac{0.007}{0.025} = 0.28 \text{ mol/dm}^3$$

Question 3

A solution containing 3.80 g of magnesium chloride is reacted with excess aqueous silver nitrate.



What is the mass of silver chloride produced?

- A 2.87 g
- B 4.98 g
- C 5.74 g
- D 11.48 g

Answer: D (11.48 g)

Explanation:

1. Molar mass of $MgCl_2$:

$$24 + (2 \times 35.5) = 24 + 71 = 95 \text{ g/mol}$$

2. Moles of $MgCl_2$:

$$\frac{3.80}{95} = 0.04 \text{ mol}$$

3. From equation: $1 \text{ mol } MgCl_2 \rightarrow 2 \text{ mol } AgCl$.

$$\text{Moles of } AgCl = 0.04 \times 2 = 0.08 \text{ mol}$$

4. Molar mass of $AgCl$:

$$108 + 35.5 = 143.5 \text{ g/mol}$$

$$\text{Mass} = 0.08 \times 143.5 = 11.48 \text{ g}$$

Question 4

5.85 g of sodium chloride (NaCl) is dissolved in a solution with a total volume of 100 cm^3 . 90 cm^3 of this solution is poured into a large beaker and the remaining 10 cm^3 into a small beaker. Which statement is correct?

- A The concentration of NaCl in the large beaker is 0.1 mol/dm^3 .
- B The concentration of NaCl in the large beaker is 52.65 g/dm^3 .
- C The concentration of NaCl in the small beaker is 0.585 g/dm^3 .
- D The concentration of NaCl in the small beaker is 1 mol/dm^3 .

Answer: D (The concentration of NaCl in the small beaker is 1 mol/dm^3)

Explanation:

1. Molar mass of NaCl = $23 + 35.5 = 58.5 \text{ g/mol}$

2. Moles of NaCl in original 100 cm^3 :

$$\frac{5.85}{58.5} = 0.1 \text{ mol}$$

3. Concentration of original solution:

$$\frac{0.1 \text{ mol}}{0.1 \text{ dm}^3} = 1 \text{ mol/dm}^3$$

4. Concentration is the same in both beakers (homogeneous solution).

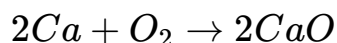
Thus, in small beaker (10 cm³): concentration = 1 mol/dm³.

Check other options:

- A: 0.1 mol/dm³ ✗ (should be 1 mol/dm³)
- B: 52.65 g/dm³ ✗ (1 mol/dm³ = 58.5 g/dm³)
- C: 0.585 g/dm³ ✗ (far too low)

Question 5

2.0 g of calcium is completely burnt in pure oxygen.



Which volume of oxygen is used at room temperature and pressure?

(Molar volume at RTP = 24 dm³/mol)

- A 0.025 dm³
- B 0.050 dm³
- C 0.60 dm³
- D 1.2 dm³

Answer: D (1.2 dm³)

Explanation:

1. Molar mass of Ca = 40 g/mol.

$$\text{Moles of Ca} = \frac{2.0}{40} = 0.05 \text{ mol}$$

2. From equation: 2 mol Ca reacts with 1 mol O₂.

$$\text{Moles of O}_2 = \frac{0.05}{2} = 0.025 \text{ mol}$$

3. Volume at RTP = moles × 24 dm³/mol:

$$0.025 \times 24 = 0.6 \text{ dm}^3$$

Correction: The answer is **C (0.60 dm³)**, not D.

(Checking your marking: C is correct – 0.6 dm³.)

Final Answer Summary:

1. D

2. A

3. D

4. D

5. C